

Forward $z^0 \rightarrow \mu^+ \mu^-$ production in pp collisions using the CDSM framework

Yan Bandeira^{1,2,*}

¹*Federal University of Pelotas, Pelotas, Brazil*

²*Institute of Nuclear Physics of the Polish Academy of Sciences, Kraków, Poland*

(Dated: March 20, 2026)

I. INTRODUCTION

Test In series of Refs. [1–3], we lay groundwork to perform a series of phenomenological work. Thus, in this document we will perform the evaluation of inclusive Z^0 production in pp collision decaying into dimuon $\mu^+\mu^-$ pair. Moreover, we will compare the numerical results with the DATA present in Ref. [4] by the LHCb collaboration.

In Ref. [4], the production cross-section was measured in the following pseudorapidity region $2.0 < \eta < 4.5$ and transverse momentum $p_T > 20$ GeV/c for both muons and dimuon invariant mass $60 < M_{\mu\mu} < 120$ GeV/c² at $\sqrt{s} = 13$ TeV.

This document has the following structure, in the next section we will approach the full process differential cross-section i.e. $pp \rightarrow (Z^0 \rightarrow \mu^+\mu^-)X$. In the third section, we will discuss only the Z^0 electroweak gauge boson production cross-section which relates with the prescription present in Refs. [1, 2] by us.

II. FULL DIFFERENTIAL CROSS-SECTION

In this section, we will discuss the forward dimuon cross-section. At forward rapidities, one expects the collinear factorization. Thus, in this scenario a hybrid factorization scheme must be used, in particular we will use the color – dipole S -matrix framework. In the hybrid factorization scheme, the lepton pair production is described by (in what follows $M \equiv M_{\mu\mu}$)

$$\frac{d\sigma(pp \rightarrow [Z^0 \rightarrow \mu^+\mu^-]X)}{d^2p_T dM^2 d\eta} = \mathcal{F}_{Z^0 \rightarrow \mu^+\mu^-}(M) \frac{d\sigma(pp \rightarrow Z^0 X)}{d^2p_T d\eta} \quad (1)$$

where

$$\mathcal{F}_{Z^0 \rightarrow \mu^+\mu^-}(M) = \text{Br}(Z^0 \rightarrow \mu^+\mu^-) \rho_Z(M). \quad (2)$$

Here, the branching ration $\text{Br}(Z^0 \rightarrow \mu^+\mu^-) \simeq 3.3662\%$, and $\rho_Z(M)$ is the invariant mass distribution of the Z^0 boson in the narrow width approximation

$$\rho_Z(M) = \frac{1}{\pi} \frac{M\Gamma_Z(M)}{(M^2 - m_Z^2)^2 + (M\Gamma(M))^2} \quad (3)$$

for

$$\frac{\Gamma_Z(M)}{M} \ll 1, \quad (4)$$

* yan.bandeira@ufpel.edu.br

in terms of the on-shell Z^0 boson mass, $m_Z \simeq 91.2$ GeV, and the generalized total Z^0 decay width

$$\Gamma_Z(M) = \frac{\alpha_{\text{em}} M}{6 \sin^2 2\theta_W} \left(\frac{160}{3} \sin^4 \theta_W - 40 \sin^2 \theta_W + 21 \right). \quad (5)$$

III. Z^0 PRODUCTION DIFFERENTIAL CROSS-SECTION

In eq. (1) r.h.s, $d\sigma(pp \rightarrow Z^0 X)$ is the inclusive Z^0 gauge boson production with invariant mass M and transverse momentum p_T in terms of the quark (antiquark) densities q_f ($\bar{q}_f$) at momentum fraction $x_q = x_1/z$ which is given by

$$\frac{d\sigma(pp \rightarrow Z^0 X)}{d^2 p_T d\eta} = J(\eta, p_T) \frac{x_1}{x_1 + x_2} \sum_f \sum_{L,T} \int_{x_1}^1 \frac{dz}{z^2} \left[q_f(x_1/z, \mu_F^2) + \bar{q}_f(x_1/z, \mu_F^2) \right] \frac{d\sigma(qp \rightarrow Z^0 X)}{d \ln z d^2 p_T} \quad (6)$$

where

$$J(\eta, p_T) \equiv \frac{dx_F}{d\eta} = \frac{2}{\sqrt{s}} \sqrt{M^2 + p_T^2} \cosh \eta, \quad (7)$$

is the Jaccobian of transformation between Feynman variable $x_F = x_1 - x_2$ and pseudorapidity η of the virtual gauge boson, Z^0 . The differential cross-section in the r.h.s of eq. (6) is the parton-target cross section which describe the in interaction between the projectile quark with the proton target producing a Z^0 gauge boson. This parton-target cross-section is derived in details in Ref. [1] which has the following form:

$$\left. \frac{d\sigma_T(q \rightarrow qZ)}{d \ln z dp_T} \right|_V = \frac{(C_f^G)^2 (g_{V,f})^2}{2\pi^2} \int dk k f(x, k) \left\{ z^4 m_f^2 \mathcal{E}_1(z, p, k, \epsilon) + [1 + (1 - z)^2] \mathcal{E}_2(z, p, k, \epsilon) \right\} \quad (8)$$

$$\left. \frac{d\sigma_T(q \rightarrow qZ)}{d \ln z dp_T} \right|_A = \frac{(C_f^G)^2 (g_{A,f})^2}{2\pi^2} \int dk k f(x, k) \left\{ z^2 m_f^2 (2 - z)^2 \mathcal{E}_1(z, p, k, \epsilon) + [1 + (1 - z)^2] \mathcal{E}_2(z, p, k, \epsilon) \right\} \quad (9)$$

$$\left. \frac{d\sigma_L(q \rightarrow qZ)}{d \ln z dp_T} \right|_V = \frac{(C_f^G)^2 (g_{V,f})^2}{4\pi^2} \int dk k f(x, k) 4(1 - z)^2 M^2 \mathcal{E}_1(z, p, k, \epsilon) \quad (10)$$

$$\left. \frac{d\sigma_L(q \rightarrow qZ)}{d \ln z dp_T} \right|_A = \frac{(C_f^G)^2 (g_{A,f})^2}{4\pi^2} \int dk k f(x, k) \left\{ 4 \frac{(z^2 m_f^2 + (1 - z) M^2)^2}{M^2} \mathcal{E}_1(z, p, k, \epsilon) + 4 \frac{z^2 m_f^2}{M^2} \mathcal{E}_2(z, p, k, \epsilon) \right\} \quad (11)$$

these four expression are the four possible polarization combination. Where $f(x, k)$ is the Unintegrated Gluon Distribution (UGD) and $\epsilon^2 = (1 - z)M^2 + z^2m_f^2$. Moreover,

$$\mathcal{E}_1(z, p, \epsilon) = \frac{1}{2} \int_0^{2\pi} d\theta \left\{ \frac{1}{[\tau^2 + \epsilon^2]^2} - \frac{2}{(p^2 + \epsilon^2)[\tau^2 + \epsilon^2]} + \frac{1}{(p^2 + \epsilon^2)} \right\}, \quad (12)$$

$$\mathcal{E}_2(z, p, \epsilon) = \frac{1}{2} \int_0^{2\pi} d\theta \left\{ \frac{\tau^2}{[\tau^2 + \epsilon^2]^2} - \frac{2\eta}{(p^2 + \epsilon^2)[\tau^2 + \epsilon^2]} + \frac{p^2}{(p^2 + \epsilon^2)^2} \right\}. \quad (13)$$

for

$$\tau^2 = (\mathbf{p} - z\mathbf{k})^2 = p^2 + z^2k^2 - 2zpk \cos \theta. \quad (14)$$

$$\eta = \mathbf{p} \cdot (\mathbf{p} - z\mathbf{k}) = p^2 - zpk \cos \theta \quad (15)$$

One can simplify the parton-target cross-section only in terms of transverse and longitudinal ones as: ($C_f^Z = \frac{\sqrt{\alpha_{\text{em}}}}{\sin 2\theta_W}$)

$$\begin{aligned} \frac{d\sigma_T(q \rightarrow qZ)}{d \ln z dp_T} &= \frac{\sqrt{\alpha_{\text{em}}}}{2\pi^2 \sin 2\theta_W} \int dk k f(x, k) \left\{ \left[(g_{V,f})^2 z^4 m_f^2 + (g_{A,f})^2 z^2 m_f^2 (2 - z)^2 \right] \mathcal{E}_1(z, p, k, \epsilon) \right. \\ &\quad \left. + \left[1 + (1 - z)^2 \right] \left((g_{V,f})^2 + (g_{A,f})^2 \right) \mathcal{E}_2(z, p, k, \epsilon) \right\} \end{aligned} \quad (16)$$

and

$$\begin{aligned} \frac{d\sigma_L(q \rightarrow qZ)}{d \ln z dp_T} &= \frac{\sqrt{\alpha_{\text{em}}}}{\pi^2 \sin 2\theta_W} \int dk k f(x, k) \left\{ (g_{A,f})^2 \frac{z^2 m_f^2}{M^2} \mathcal{E}_2(z, p, k, \epsilon) \right. \\ &\quad \left. + \left[(g_{V,f})^2 (1 - z)^2 M^2 + (g_{A,f})^2 \frac{(z^2 m_f^2 + (1 - z)M^2)^2}{M^2} \right] \mathcal{E}_1(z, p, k, \epsilon) \right\} \end{aligned} \quad (17)$$

defining:

$$\Gamma_T = (g_{V,f})^2 z^4 m_f^2 + (g_{A,f})^2 z^2 m_f^2 (2 - z)^2 \quad (18)$$

$$\Gamma_L = (g_{V,f})^2 (1 - z)^2 M^2 + (g_{A,f})^2 \frac{(z^2 m_f^2 + (1 - z)M^2)^2}{M^2} \quad (19)$$

$$\Lambda_T = \left[1 + (1 - z)^2 \right] \left((g_{V,f})^2 + (g_{A,f})^2 \right) \quad (20)$$

$$\Lambda_L = (g_{A,f})^2 \frac{z^2 m_f^2}{M^2} \quad (21)$$

Thus, the parton–target cross-section can be reduce to one expression:

$$\frac{d\sigma(q \rightarrow qZ)}{d \ln z dp_T} = \frac{\sqrt{\alpha_{\text{em}}}}{2\pi^2 \sin 2\theta_W} \int dk k f(x, k) \left\{ \left(\Gamma_T + 2\Gamma_L \right) \mathcal{E}_1(z, p, k, \epsilon) + \left(\Lambda_T + 2\Lambda_L \right) \mathcal{E}_2(z, p, k, \epsilon) \right\} \quad (22)$$

IV. IMPACT PARAMETER SPACE

The four parton-level cross sections are, based on [1] but with $m_a = m_b$:

$$\left. \frac{d\sigma_T(q \rightarrow qZ)}{d \ln z dp_T} \right|_V = \frac{(C_f^G)^2 (g_{V,f})^2}{2\pi^2} \left\{ z^4 m_f^2 \mathcal{D}_1(z, p, \epsilon) + [1 + (1 - z)^2] \epsilon^2 \mathcal{D}_2(z, p, \epsilon) \right\} \quad (23)$$

$$\left. \frac{d\sigma_T(q \rightarrow qZ)}{d \ln z dp_T} \right|_A = \frac{(C_f^G)^2 (g_{A,f})^2}{2\pi^2} \left\{ z^2 m_f^2 (2 - z)^2 \mathcal{D}_1(z, p, \epsilon) + [1 + (1 - z)^2] \epsilon^2 \mathcal{D}_2(z, p, \epsilon) \right\} \quad (24)$$

$$\left. \frac{d\sigma_L(q \rightarrow qZ)}{d \ln z dp_T} \right|_V = \frac{(C_f^G)^2 (g_{V,f})^2}{4\pi^2} 4(1 - z)^2 M^2 \mathcal{D}_1(z, p, k, \epsilon) \quad (25)$$

$$\left. \frac{d\sigma_L(q \rightarrow qZ)}{d \ln z dp_T} \right|_A = \frac{(C_f^G)^2 (g_{A,f})^2}{4\pi^2} \left\{ 4 \frac{(z^2 m_f^2 + (1 - z) M^2)^2}{M^2} \mathcal{D}_1(z, p, \epsilon) + 4 \frac{z^2 m_f^2}{M^2} \mathcal{D}_2(z, p, \epsilon) \right\} \quad (26)$$

collecting the transverse contributions:

$$\begin{aligned} \frac{d\sigma_T(q \rightarrow qZ)}{d \ln z dp_T} &= \frac{(C_f^G)^2}{2\pi^2} \left\{ \left(g_{V,f}^2 z^4 m_f^2 + g_{A,f}^2 z^2 m_f^2 (2 - z)^2 \right) \mathcal{D}_1(z, p, \epsilon) \right. \\ &\quad \left. + \left(g_{V,f}^2 + g_{A,f}^2 \right) [1 + (1 - z)^2] \epsilon^2 \mathcal{D}_2(z, p, \epsilon) \right\} \end{aligned} \quad (27)$$

for the longitudinal contribution:

$$\frac{d\sigma_L(q \rightarrow qZ)}{d \ln z dp_T} = \frac{(C_f^G)^2}{\pi^2} \left\{ g_{A,f}^2 \epsilon^2 \frac{z^2 m_f^2}{M^2} \mathcal{D}_2(z, p, \epsilon) \right. \quad (28)$$

$$\left. + \left[g_{V,f}^2 (1 - z)^2 M^2 g_{A,f}^2 + \frac{(z^2 m_f^2 + (1 - z) M^2)^2}{M^2} \right] \mathcal{D}_1(z, p, \epsilon) \right\} \quad (29)$$

defining:

$$\Gamma_T = (g_{V,f})^2 z^4 m_f^2 + (g_{A,f})^2 z^2 m_f^2 (2-z)^2 \quad (30)$$

$$\Gamma_L = (g_{V,f})^2 (1-z)^2 M^2 + (g_{A,f})^2 \frac{(z^2 m_f^2 + (1-z)M^2)^2}{M^2} \quad (31)$$

$$\Lambda_T = \epsilon^2 [1 + (1-z)^2] \left((g_{V,f})^2 + (g_{A,f})^2 \right) \quad (32)$$

$$\Lambda_L = \epsilon^2 (g_{A,f})^2 \frac{z^2 m_f^2}{M^2} \quad (33)$$

Thus, the parton–target cross-section can be reduce to one expression:

$$\frac{d\sigma(q \rightarrow qZ)}{d \ln z dp_T} = \frac{(C_f^G)^2}{2\pi^2} \left\{ \left(\Gamma_T + 2\Gamma_L \right) \mathcal{D}_1(z, p, \epsilon) + \left(\Lambda_T + 2\Lambda_L \right) \mathcal{D}_2(z, p, \epsilon) \right\} \quad (34)$$

-
- [1] Y. B. Bandeira, V. P. Goncalves, and W. Schäfer, [JHEP **07**, 171, arXiv:2405.10265 \[hep-ph\]](#).
 - [2] Y. B. Bandeira, V. P. Goncalves, and W. Schäfer, [Phys. Rev. D **111**, 074041 \(2025\), arXiv:2411.12675 \[hep-ph\]](#).
 - [3] Y. B. Bandeira, V. P. Goncalves, and W. Schäfer, (2025), [arXiv:2507.06207 \[hep-ph\]](#).
 - [4] R. Aaij *et al.* (LHCb), [JHEP **07**, 026, arXiv:2112.07458 \[hep-ex\]](#).